COMMERCIAL ITEM DESCRIPTION

WIRE, WELDING, HARD FACING, COBALT BASE ALLOY

The General Services Administration has Authorized the use of this commercial item description.

Abstract. This description covers cobalt welding wire of the standard, commercial type, which is for use with the inert gas shielded arc welding process for hard facing of chromium-tungsten-nickel-cobalt base metal.

Classification. The cobalt welding wire shall be of the following types and sizes. The type and size to be furnished shall be as specified (see ordering data).

Type I - Cut length

Size 1	-	0.032 inch diameter	± 0.005 in.
Size 2	-	0.045 inch diameter	± 0.005 in.
Size 3	-	0.065 inch diameter	± 0.005 in.
Size 4	-	0.094 inch diameter	± 0.005 in.
Size 5	-	0.125 inch diameter	± 0.005 in.

Type II - Spools

Size 1	-	0.032 inch diameter	± 0.005 in.
Size 2	-	0.045 inch diameter	± 0.005 in.
Size 3	-	0.065 inch diameter	± 0.005 in.

Salient characteristics.

1. Design. The metal shall be of cast form then hot extruded into wire and centerless ground. The wire ends may be butt welded in order to provide a continuous wire, provided the metal is from the same heat. The welded joint shall have a tolerance of ± 0.005 inches of the original wire diameter and shall provide an uninterrupted feed of the wire when used with inert gas semiautomatic and automatic welding equipment.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5051.

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2. Wire length. Unless otherwise specified (see ordering data), the type I wire shall be furnished in lengths of 18 or 24 inches. Type II wire shall be furnished in continuous length and unless otherwise specified(see ordering data), shall be wound on 12 inch spools which meet the requirements of figure 1, AWS A5.14.

3. Chemical composition. The wire shall be furnished with the following chemical composition.

TABLE I. Chemical composition.

Composition	Minimum	Maximum
Carbon	0.70	1.00
Manganese	--	1.00
Silicon	--	1.00
Chromium	26.00	30.00
Nickel	4.00	6.00
Tungsten	18.00	21.00
Boron	0.005	0.10
Vanadium	0.75	1.25
Iron	--	3.00
Molybdenum	--	0.50
Cobalt	Remainder	

4. Weldability. The hard facing wire shall flow freely leaving a weld deposit free of porosity, gas cavities, and slag inclusions when welded by the shielded inert gas process. The weld deposit shall have a hardness not less than a Rockwell 50 on the "C" scale.

5. Identification. Each roll or package of wire shall be furnished with the following information:

Manufacturers name
Batch and heat number
Diameter of wire

Each length of type 1 wire shall be tagged with a flag tag which states the following "PWA-694".

6. Workmanship. Workmanship shall provide a wire having a smooth finish, free from slivers, depressions, scale or other foreign matter that would adversely affect the welding characteristics or properties of the weld metal.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and

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quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimension will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

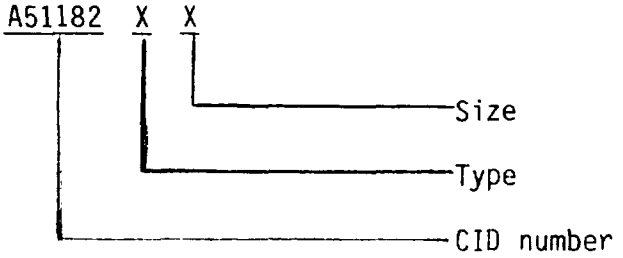
If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, and marking. Unless otherwise specified in the contract or purchase order preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see ordering data), preservation, packing, and marking shall be in accordance with MIL-W-10430 level as specified in the contract.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.



Quality assurance.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Sampling inspection. As a minimum sampling inspection, the supplier shall select a test sample from each heat furnished. Each test sample shall be inspected to determine conformance with the examination, the chemical analysis, the dimensional inspection, and the hardness test. Should any of the sample items fail to meet either the examination, the chemical analysis, the inspection, or the test, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items of the same size presented at one time for acceptance.

Examination. Each sample of wire selected shall be visually examined to determine conformance with paragraphs 5 and 6 of the salient characteristics.

Dimensional test. The wire diameter shall be tested for conformance with the dimensional requirements of paragraphs 1 and 2 of the salient characteristics.

Chemical analysis. A chemical analysis of the sample(s) shall be performed in accordance with ASTM E 1019, E 1085, or E 1086 to determine conformance with paragraph 3 of the salient characteristics.

Weldability test. A weldability test shall be performed in accordance with paragraph 2 thru 2.6.2, SAE ARP 1876 to determine conformance with paragraph 4 of the salient characteristics. Unless otherwise specified in the ordering data, a low carbon steel test specimen, 0.060 inch thick, 10 inches in length and 2 inches in width shall be used to perform the test.

Hardness test. A hardness test shall be performed in accordance with ASTM E 18 to determine conformance with the hardness requirements of paragraph 4 of the salient characteristics.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Type, size, and quantity required.

c. Length wire required.

d. Describe spool, if different.

d. Packaging in accordance with MIL-W-10430 if required; specify level.

e. Test specimen requirements, if different.

Federal Standard FED-STD-376 is available from the Naval Publications and Form center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

Military Specification MIL-W-10430 is available from the Naval Publications and Form center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

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ASTM E 1019, E 1085, E 1086, and D 3951 are available from the American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

AWS A5.14, is available from the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135-0440.

SAE ARP 1876 is available from the Customer Service Department, Publications Group, SAE, 400 Commonwealth Drive, Warrendale, PA 15096-7511.

Definitions.

Batch. The combining and mixing of metals into one single unit.

Heat. The single melting of alloy in a single furnace.

MILITARY INTEREST:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activity

DLA - GS

CIVIL AGENCY COORDINATION ACTIVITY:

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PREPARING ACTIVITY:

DLA - IP

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